ELAKURTHI PAVANKUMAR

AI/ML Engineer | Machine Learning | Computer Vision | Data Analytics

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PROFESSIONAL SUMMARY

Aspiring AI/ML Engineer with hands-on experience in machine learning, computer vision, and data analytics using Python. Skilled in building data-driven solutions including recommendation systems, gesture-based computer vision applications, and predictive models using Scikit-learn, Pandas, NumPy, and OpenCV. Experienced in data preprocessing, exploratory data analysis (EDA), and end-to-end model development on real-world datasets. Passionate about applying artificial intelligence to solve practical problems.

TECHNICAL SKILLS

Programming Languages: Python (Pandas, NumPy, Matplotlib)

ML/DL Libraries: Scikit-learn, TensorFlow/Keras, OpenCV, MediaPipe, BeautifulSoup

Machine Learning: Supervised Learning, Unsupervised Learning, Regression, Classification, Clustering, Recommendation Systems

Deep Learning: Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), Transfer Learning

Databases: MySQL, SQL (Joins, Aggregations, Subqueries, Transactions)

Data Analysis: Data Cleaning, Exploratory Data Analysis (EDA), Feature Engineering, Data Visualization

Visualization Tools: Power BI, Matplotlib, Excel

Tools & Platforms: Jupyter Notebook, Google Colab, Git, GitHub

PROJECTS

AI Virtual Mouse using Computer Vision | GitHub Repository

- Developed a real-time virtual mouse system using Python, OpenCV, and MediaPipe for hand gesture recognition.
- Implemented hand tracking algorithms to control cursor movement and perform mouse actions such as left-click, right-click, and scrolling.
- Optimized frame processing pipeline to improve responsiveness and achieve smooth real-time performance.

Netflix Movie Recommendation System | GitHub Repository

- Built a content-based filtering movie recommendation system using Python and cosine similarity scoring.
- Processed and cleaned movie metadata, and applied feature extraction techniques (TF-IDF, vectorization) to generate personalized movie recommendations.
- Evaluated recommendation quality and implemented similarity scoring to improve recommendation accuracy.

Netflix Data Analysis and Visualization | GitHub Repository

- Performed comprehensive exploratory data analysis (EDA) on Netflix datasets using Python, Pandas, and Matplotlib.
- Created visualizations to analyze trends in content genres, ratings distribution, and release year patterns.
- Extracted actionable insights from data to understand content distribution and platform growth patterns.

Global Energy Consumption Analysis | GitHub Repository

- Analyzed global energy consumption datasets using advanced SQL queries and data aggregation techniques.
- Applied complex joins, filtering, and grouping operations to explore energy usage patterns across multiple countries and time periods.
- Generated analytical insights on historical energy consumption trends and cross-country comparisons.

PROFESSIONAL TRAINING

Data Science Training Program | Innomatics Research Labs — Hyderabad • Present

- Learning Python for Data Science, Machine Learning, and Data Analysis with structured curriculum.
- Hands-on experience with Pandas, NumPy, Scikit-learn, and industry-standard data visualization tools.
- Working on real-world projects involving end-to-end data analysis and machine learning model development.

CERTIFICATIONS

- Python for Data Science – Innomatics Research Labs (View Certificate)
- SQL for Data Analysis – Innomatics Research Labs (View Certificate)
- Exploratory Data Analysis (EDA) – Innomatics Research Labs (View Certificate)

EDUCATION

Bachelor of Computer Applications (BCA) — 2025

Nizam College, Hyderabad CGPA: 8.69 / 10

Intermediate – MPC — 2022

Alphores Junior College, Telangana State Board CGPA: 9.8 / 10

Secondary School Certificate (SSC) — 2020

Saraswathi Vidhya Vihar High School, Telangana State Board CGPA: 10 / 10